WTW158 CALCULUS
SEMESTER TEST 1 / SEMESTERTOETS 1

Date / Datum: March /Maart 2018   Marks / Punte : 45   Time / Tyd : $2\frac{1}{2}$ hours / uur

MARKS FOR SECTIONS B & C PUNTE VIR AFDELINGS B & C	
	30

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SURNAME / VAN | M | E | M | O | | | | | | | | | | | |

FIRST NAMES / VOORNAME ____________________

STUDENT NUMBER
STUDENTENOMMER | | | | | | | | |

SIGNATURE
HANDTEKENING ____________________

CONTACT NUMBER
KONTAKNOMMER | | | | | | | | | | | |

CIRCLE YOUR LECTURE GROUP NUMBER
OMKRING JOU LESINGGROEPNOMMER.

1	2	3	4	10
Dr Jooste	Prof Harding	Dr Garba	Dr van der Hoff	Ms Dreyer

READ THE FOLLOWING INSTRUCTIONS
1. The paper consists of pages 1 to 10. Check whether your paper is complete.
2. **No calculator may be used.**
3. No question paper may be removed from the venue.

LEES DIE VOLGENDE INSTRUKSIES
1. Die vraestel bestaan uit bladsye 1 tot 10. Kontroleer of u vraestel volledig is.
2. **Geen sakrekenaar mag gebruik word nie.**
3. Geen vraestel mag uit die lokaal geneem word nie.

Section A: / Afdeling A:

Marks: 15 / Punte: 15 (1.5 per question / vraag)

1. The responses to questions in Section A must be completed on SIDE 2 of the optical reader form in SOFT PENCIL.
2. First circle your answers on this paper and then transfer these to the optical reader form.
3. There is only ONE correct answer per question.

1. Die antwoorde op vrae in Afdeling A moet op KANT 2 van die merkleesvorm met 'n SAGTE POTLOOD voltooi word.
2. Omkring eers jou antwoorde op hierdie vraestel en dra dit dan oor na die merkleesvorm.
3. Daar is slegs EEN korrekte antwoord per vraag.

Question 1 / Vraag 1

The solution set of $\dfrac{|x-2|}{\sqrt{x}} > 0$ is

Die oplossingsversameling van $\dfrac{|x-2|}{\sqrt{x}} > 0$ is

(a)	$(-\infty,2)\cup(2,\infty)$	(b)	$(0,\infty)$	(c)	$(0,2)$
(d)	$(-2,\infty)$	(e)	$(0,\infty)\backslash\{2\}$	(f)	None of these / Geen van hierdie

Question 2 / Vraag 2

The solution set of $4-|3-x| < 2$ is

Die oplossingsversameling van $4-|3-x| < 2$ is

(a)	$(-\infty,1)\cup(5,\infty)$	(b)	$(-5,1)$	(c)	$(1,5)$
(d)	$(-\infty,-5)\cup(1,\infty)$	(e)	$(1,5)\backslash\{3\}$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

The domain of the function $f(x)=\sqrt{1-9x^2}$ is

Die definisieversameling van die funksie $f(x)=\sqrt{1-9x^2}$ is

(a)	$[3,\infty)$	(b)	$(-\infty,-3]\cup[3,\infty)$	(c)	$(-\infty,-\frac{1}{3}]\cup[\frac{1}{3},\infty)$
(d)	$[-\frac{1}{3},\frac{1}{3}]$	(e)	$(-\infty,-\frac{1}{3}]$ or $(-\infty,\frac{1}{3}]$	(f)	None of these / Geen van hierdie

Question 4 / Vraag 4

The range of the function $f(x) = \frac{|x|}{x}$ is

Die waardeversameling van die funksie $f(x) = \frac{|x|}{x}$ is

(a)	$\{-1, 1\}$	(b)	$[1, \infty)$	(c)	$(-\infty, 1]$
(d)	$(0, \infty)$	(e)	$\mathbb{R}\backslash\{0\}$	(f)	None of these / Geen van hierdie

Question 5 / Vraag 5

If / Indien $\sec\theta = -\frac{2}{\sqrt{3}}$ and / en $0 \le \theta < 2\pi$, then / dan is $\theta =$

(a)	$\frac{2\pi}{3}$ or / of $\frac{4\pi}{3}$	(b)	$\frac{5\pi}{4}$ or / of $\frac{7\pi}{4}$	(c)	$\frac{7\pi}{6}$ or / of $\frac{11\pi}{6}$
(d)	$\frac{5\pi}{6}$ or / of $\frac{7\pi}{6}$	(e)	$\frac{4\pi}{3}$ or / of $\frac{5\pi}{3}$	(f)	None of these / Geen van hierdie

Question 6 / Vraag 6

The value of $\cos(\frac{23\pi}{3})$ is / Die waarde van $\cos(\frac{23\pi}{3})$ is

(a)	$-\frac{\sqrt{3}}{2}$	(b)	$\frac{\sqrt{3}}{2}$	(c)	$\frac{1}{2}$
(d)	$-\frac{1}{2}$	(e)	1	(f)	None of these / Geen van hierdie

Question 7 / Vraag 7

f is an **odd** function defined on $\mathbb{R}$. On $(0, \infty)$ it is defined as
f is 'n **onewe** funksie gedefinieer op $\mathbb{R}$. Op $(0, \infty)$ word dit gedefinieer deur

$$f(x) = \begin{cases} 2 & \text{if } 0 < x \le 1 \\ 2 + x & \text{if } x > 1 \end{cases}$$

The value of $f(-4)$ is / Die waarde van $f(-4)$ is

(a)	2	(b)	-2	(c)	6
(d)	-6	(e)	-4	(f)	None of these / Geen van hierdie

Question 8 / Vraag 8

The solution of $e^{x-2}(x^2 - x) \leq 0$ is

Die oplossing van $e^{x-2}(x^2 - x) \leq 0$ is

(a)	$x \in [1, \infty)$	(b)	$x \in (-\infty, 0] \cup [1, \infty)$	(c)	$x \in [0, 1]$
(d)	$x \in [-\infty, 1]$	(e)	$x \in \mathbb{R}$	(f)	None of these / Geen van hierdie

Question 9 / Vraag 9

Given the function / Gegee die funksie $f(x) = \begin{cases} -x + 2 & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$

$(f \circ f)(3) =$

(a)	1	(b)	3	(c)	Undefined / Ongedefinieerd
(d)	0	(e)	$\frac{1}{e}$	(f)	None of these / Geen van hierdie

Question 10 / Vraag 10

The average rate of change of $f(x) = \frac{1}{x}$ over the interval $[1, 3]$ is

Die gemiddelde tempo van verandering van $f(x) = \frac{1}{x}$ oor die interval $[1, 3]$ is

(a)	-2	(b)	$\frac{1}{3}$	(c)	$-\frac{2}{3}$
(d)	$-\frac{1}{3}$	(e)	$-\frac{1}{9}$	(f)	None of these / Geen van hierdie

Section B Marks: 12 / Afdeling B Punte: 12

1. This section must be completed IN PEN on this paper (including sketches).
2. Fill in the answer only in the space provided. Rough work can be done on the facing page and will not be marked.

1. Hierdie afdeling moet MET PEN op hierdie vraestel gedoen word (insluitend sketse).
2. Vul slegs die antwoord in die spasie in. Rofwerk kan op die teenblad gedoen word en sal nie gemerk word nie.

Question 11 / Vraag 11

11.1 Write the function $f(x) = \ln|x-1|$ as a piece-wise defined function.

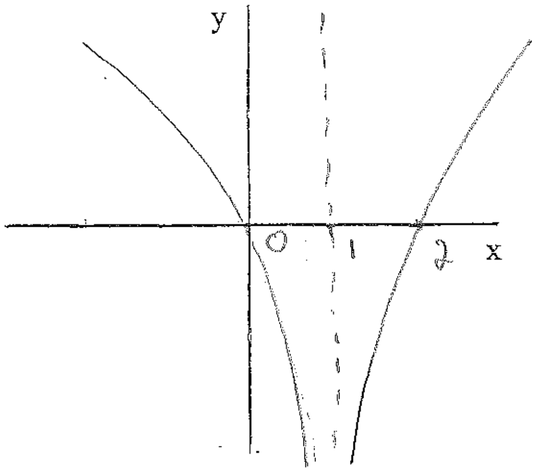
Skryf die funksie $f(x) = \ln|x-1|$ as 'n stuksgewys gedefinieerde funksie.

$$f(x) = \begin{cases} \ln(x-1) & \text{if / as } x \in (1, \infty) \\ \ln(1-x) & \text{if / as } x \in (-\infty, 1) \end{cases}$$

2

11.2 Sketch $f(x) = \ln|x-1|$.

Skets $f(x) = \ln|x-1|$.



2

11.3 Use the graph in 11.2 to solve: $\ln|x-1| \leq 0$

Gebruik die grafiek in 11.2 om op te los: $\ln|x-1| \leq 0$

$x \in [0, 2] \setminus \{1\}$

1

Question 12 / Vraag 12

Expand the following expression completely: / Brei die volgende uitdrukking volledig uit:

$$\ln \frac{\sqrt{e^{x-2}}\,(x-2)^2}{e^x + e^2}$$

$\frac{1}{2}(x-2) + 2\ln(x-2) - \ln(e^x + e^2)$	2

Question 13 / Vraag 13

The function f passes through the point $(1,2)$. Give a corresponding point that the function $g(x) = 2f(x-2) - 1$ passes through.

Die funksie f gaan deur die punt $(1,2)$. Gee 'n ooreenstemmende punt waardeur die funksie $g(x) = 2f(x-2) - 1$ gaan.

$(3,3)$	1

Question 14 / Vraag 14

$\sin^{-1}(\sin(\frac{7\pi}{4})) =$

$-\pi/4$	1

Question 15 / Vraag 15

The solution of / Die oplossing van $\ln 3x = \ln x^3 + \ln \frac{3}{x}$ is

$x = 1$	1

Question 16 / Vraag 16

Given / Gegee: $f(x) = \begin{cases} \frac{1}{x-1} & \text{if } x < 1 \\ \frac{1}{e^{x-1}} & \text{if } x > 1 \end{cases}$

Complete / Voltooi:

$\lim_{x \to 1^+} f(x) =$ 1	
$\lim_{x \to 1^-} f(x) =$ $-\infty$	2

Section C Marks: 18 / Afdeling C Punte: 18

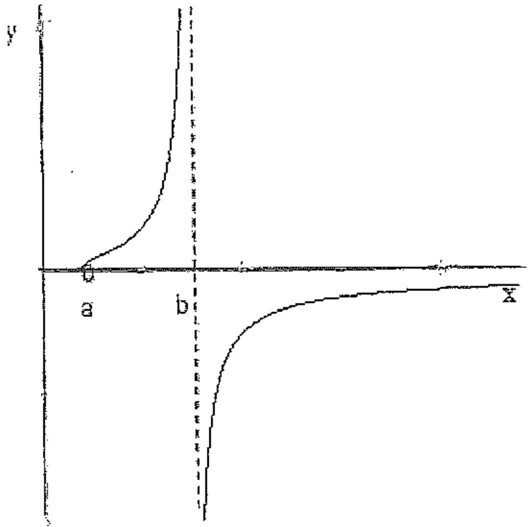
1. The responses to questions in Section C must be completed IN PEN on this paper (including sketches).
2. All calculations must be shown and will be marked.
3. For more than the available space, use the facing page and clearly indicate it.
4. Attention should be paid to writing in a mathematically sound way.

1. Die antwoorde op vrae in Afdeling C moet MET PEN op die vraestel voltooi word (insluitend sketse).
2. Alle berekeninge moet getoon word en sal gemerk word.
3. Vir meer as die beskikbare ruimte, gebruik die teenblad en dui dit duidelik aan.
4. Maak seker dat jou skryfwyse wiskundig korrek is.

Question 17 / Vraag 17

The function $f(x) = \frac{1}{1-\ln(x-1)}$ is shown in the sketch.

Die funksie $f(x) = \frac{1}{1-\ln(x-1)}$ word getoon in die skets.



17.1 Find the value of a.

Bepaal die waarde van a.

$\ln(x-1)$ is defined for $x-1>0 \Rightarrow x>1$

$\Rightarrow a=1$

1

17.2 Find the value of b.

Bepaal die waarde van b.

$1-\ln(x-1) \neq 0$

$\Rightarrow \ln(x-1) \neq 1$

$\Rightarrow x-1 \neq e$

$\Rightarrow x \neq 1+e$

$\Rightarrow b = 1+e$

2

17.3 Is the function f one-to-one? Explain by referring to the graph of f.

Is die funksie een-een-duidig? Verduidelik deur na die grafiek van f te verwys.

Yes, it passes both the vertical and the horizontal line tests. 1

17.4 Find the inverse function f^{-1}.

Bepaal die inverse funksie f^{-1}.

$$y = \frac{1}{1-\ln(x-1)}$$

For inverse: $x = \frac{1}{1-\ln(y-1)}$

$$\Rightarrow 1-\ln(y-1) = \frac{1}{x}$$

$$\Rightarrow -\ln(y-1) = \frac{1}{x} - 1$$

$$\Rightarrow \ln(y-1) = 1 - \frac{1}{x}$$

$$\Rightarrow y-1 = e^{1-\frac{1}{x}}$$

$$\Rightarrow y = 1 + e^{1-\frac{1}{x}}$$

2

17.5 Use the graph given at the start of the question to determine the domain and range of the **inverse function** f^{-1}.

Gebruik die grafiek wat aan die begin van die vraag gegee word om die definisie- en waardeversameling van die **inverse funksie** f^{-1} te bepaal.

Domain: $\mathbb{R}\setminus\{0\}$

Range: $(1,\infty)\setminus\{1+e\}$ or $(a,\infty)\setminus\{b\}$ 2

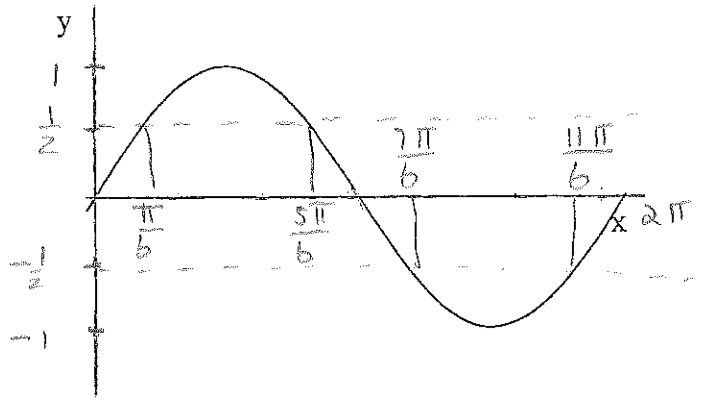
Question 18 / Vraag 18

18.1 Complete the theorem: / Voltooi die stelling:

If / As $a > 0$ then / dan is $|x| \leq a \Leftrightarrow$ $-a \leq x \leq a$ 1

18.2 The function $y = \sin x$ is given in the sketch on the interval $[0, 2\pi]$. Use the theorem in 18.1 and the sketch to solve for x on $[0, 2\pi]$ such that $|\sin x| < \frac{1}{2}$. Indicate all applicable values on the sketch.

Die funksie $y = \sin x$ word gegee in die skets op die interval $[0, 2\pi]$. Gebruik die stelling in 18.1 en die skets om vir x op te los op $[0, 2\pi]$ sò dat $|\sin x| < \frac{1}{2}$. Dui alle toepaslike waardes op die skets aan.



$-\frac{1}{2} < \sin x < \frac{1}{2} \Rightarrow x \in [0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, \frac{7\pi}{6}) \cup (\frac{11\pi}{6}, 2\pi]$

2

Question 19 / Vraag 19

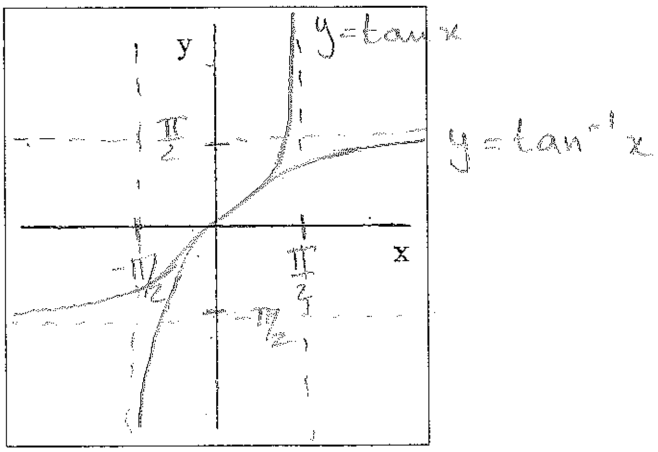
19.1 Complete the definition: / Voltooi die definisie:

$y = \tan^{-1}x \Leftrightarrow$ $\tan y = x$ and / en $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$(\tan^{-1}x = \arctan x = bg\tan x)$ 1

19.2 Sketch both the function $y = \tan x, x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and the function $y = \tan^{-1}x$ on the given set of axes. Show all the axes-intercepts and asymptotes.

Skets beide die funksie $y = \tan x$ en die funksie $y = \tan^{-1}x, x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ op die gegewe assestelsel. Toon alle assesnydings en asimptote.



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Question 20 / Vraag 20

Given the function / Gegee die funksie $f(x) = \begin{cases} -e^{-x} & \text{if/as } x < 0 \\ 1 - 2e^{x} & \text{if/as } x > 0 \end{cases}$.

20.1 Determine the following (if it exists)
Bepaal die volgende (indien dit bestaan)

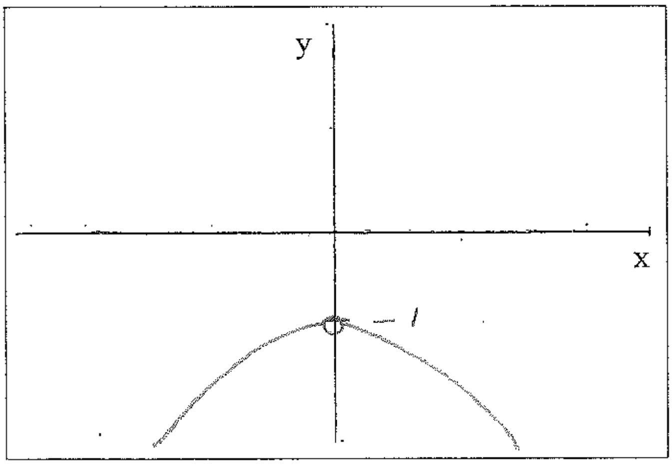
(i) $f(0)$ Does not exist / is undefined

(ii) $\lim\limits_{x\to 0} f(x)$

$$\left.\begin{aligned} \lim_{x\to 0^+} f(x) &= \lim_{x\to 0^+}(1-2e^{x}) = -1 \\ \lim_{x\to 0^-} f(x) &= \lim_{x\to 0^-}(-e^{-x}) = -1 \end{aligned}\right\} \Rightarrow \lim_{x\to 0} f(x) = -1$$

2

20.2 Sketch f / Skets f



2

Bonus question / vraag: Question 21 / Vraag 21

Show that the function $f(x) = \ln(x + \sqrt{1+x^2})$ is an odd function by using the definition of an odd function.

Wys dat die funksie $f(x) = \ln(x + \sqrt{1+x^2})$ 'n onewe funksie is deur gebruik te maak van die definisie van 'n onewe funksie.

For an odd function: $f(-x) = -f(x)$

$\Rightarrow f(-x) + f(x) = 0$ 1

LHS: $\ln(-x + \sqrt{1+x^2}) + \ln(x + \sqrt{1+x^2})$

$= \ln(-x^2 + 1 + x^2)$

$= \ln 1$

$= 0$

$=$ RHS